Matthew Voynovich - Predicting NBA Shot Outcomes using Spatial Data and Generic Player Characteristics

Progress to Date

The team has finished almost all of the technical/code components of the NBA Shot Outcome Prediction project. We have a collaborative development environment set up on a shared Jupyter Notebook in Google Colab. Additionally, we have created a GitHub repository for version control.

On the technical side, we combined the raw NBA shot-log data with the player characteristics in which we then dropped identifying information about the player (such as name and team). We performed extensive exploratory data analysis to examine feature distributions, identify outliers, and assess multicollinearity using Variance Inflation Factor (VIF). We used this information to drop highly correlated features and then performed feature engineering with some created features. Data was then split via train test split into 3 stratified groupings (training, validation, and testing. 70/15/15 split respectively) and class imbalances were addressed using SMOTE and class-weight adjustments.

First we trained a baseline logistic regression model to serve as a reference point for evaluation. Next we implemented more advanced ensemble models (like Random Forest and XGBoost) and a feed-forward neural network. All three models were tuned using random search with cross validation for hyperparameters. They were then compared on accuracy, precision, recall, and F1-score. Using these metrics, we selected our best model (currently the tuned XGBoost). All code, data pipelines, experiment logs, and preliminary results are fully documented and reproducible in the shared notebook on colab.

Current Results:

The baseline logistic regression provided a solid starting benchmark. Ensemble methods (particularly XGBoost) and the neural network both outperformed the baseline, with the tuned XGBoost model currently showing the strongest overall metrics on the test set. Detailed numerical results and confusion matrices are stored in the shared notebook.

Next Steps:

The remaining work focuses on interpretation, visualization, and final deliverables. We will generate feature importance plots (including SHAP values) and create spatial heatmaps of expected shot value across the court. All experimental results will be compiled into publication-quality visualizations such as ROC curves, precision-recall curves, and model comparison charts.